



Recycle Speed Analysis and Simulations Harlequins, Season 2020/21

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Summary

Competition: Premiership 2021
Rucks Analysed: 429
Opponents: Wasps, Leicester, Bath, Bristol, Exeter

General Tendencies for Rucks Leading to Ground Passes:

Rucks 310
Median speed 2.66 sec
Contested rucks 138 (44.5%)
LQB rucks 174 (56%)
Players committed 2 (median)

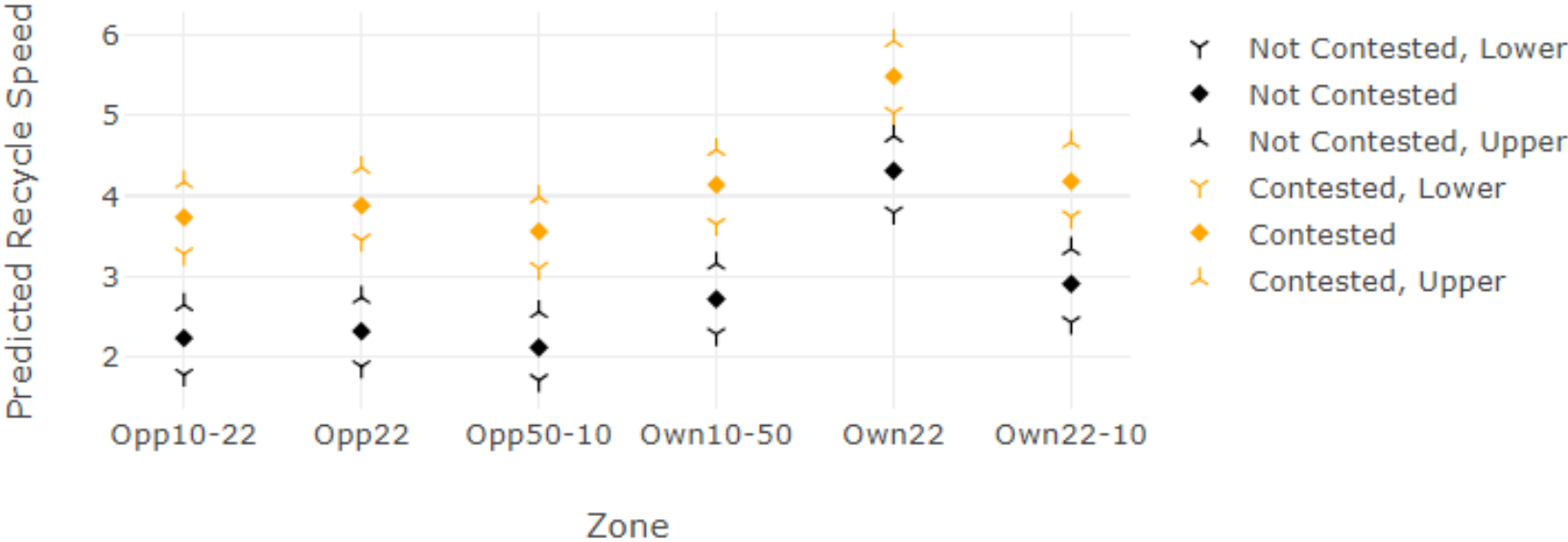
How Contesting the Ruck Influenced the Recycle Speed:

Zone	Not Contested	Contested	Difference
Own22	4.34 sec	5.57 sec	1.23 sec
Own22-10	2.86 sec	4.22 sec	1.36 sec
Own10-50	2.65 sec	4.15 sec	1.50 sec
Opp50-10	2.05 sec	3.56 sec	1.51 sec
Opp10-22	2.19 sec	3.77 sec	1.58 sec
Opp22	2.25 sec	3.90 sec	1.65 sec

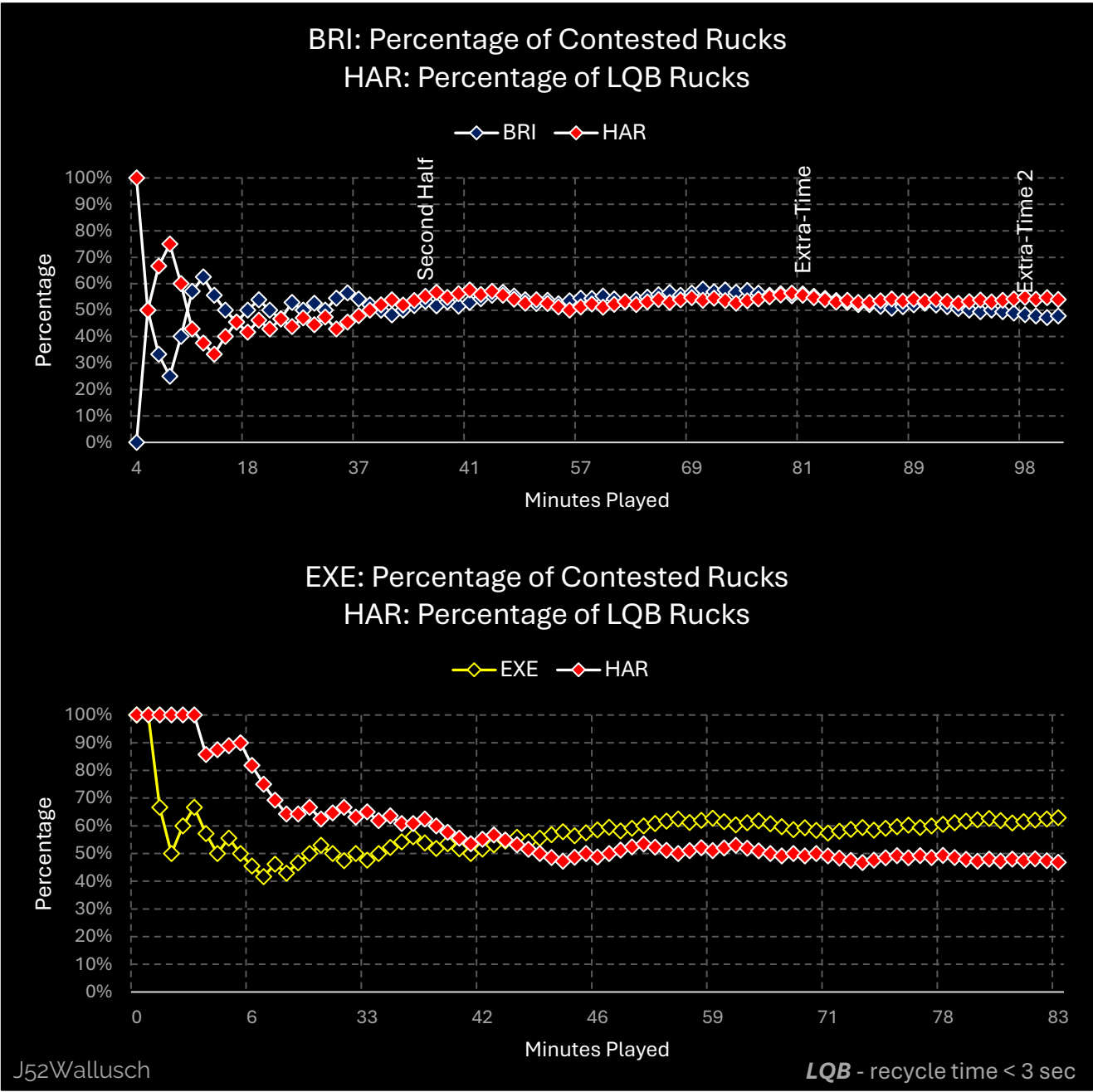
Contesting rucks **can slow down** the recycle speed by
more than 1 second



Harlequins: Simulated Impact of Counterrucking on Recycle Speed



Simulations
performed using a
machine-learning
model





Players Committed to Attacking Rucks

Number of Rucks with Players Committed					
	1	2	3	4	5
Own22	1	7	12	11	12
Own22-10	2	15	17	15	6
Own10-50	4	15	17	4	0
Opp50-10	7	23	12	2	0
Opp10-22	13	70	24	4	1
Opp22	22	70	37	6	0
	49 (11%)	200 (47%)	119 (28%)	42 (10%)	19 (4%)

HARvWAS		1	2	3	4	5
	Own22	0	2	1	1	2
	Own22-10	1	3	6	4	2
	Own10-50	0	3	5	2	0
	Opp50-10	1	1	0	0	0
	Opp10-22	1	13	1	0	1
	Opp22	5	17	6	0	0
HARvLEI		8 (10%)	39 (50%)	19 (24%)	7 (9%)	5 (6%)
	Own22	0	1	7	3	2
	Own22-10	1	5	2	4	0
	Own10-50	3	1	2	1	0
	Opp50-10	2	6	1	0	0
	Opp10-22	2	19	6	1	0
	Opp22	6	12	11	3	0
HARvBTH		14 (14%)	44 (44%)	29 (29%)	12 (12%)	2 (2%)
	Own22	0	3	2	3	1
	Own22-10	0	2	3	3	0
	Own10-50	0	2	2	0	0
	Opp50-10	0	4	1	0	0
	Opp10-22	8	19	6	1	0
	Opp22	5	8	5	0	0
HARvBRI		13 (17%)	38 (49%)	19 (24%)	7 (9%)	1 (1%)
	Own22	0	1	2	2	3
	Own22-10	0	1	1	3	0
	Own10-50	1	7	7	1	0
	Opp50-10	2	7	7	1	0
	Opp10-22	1	8	8	1	0
	Opp22	4	17	5	1	0
HARvEXE		8 (9%)	41 (45%)	30 (33%)	9 (10%)	3 (3%)
	Own22	1	0	0	2	4
	Own22-10	0	4	5	1	4
	Own10-50	0	2	1	0	0
	Opp50-10	2	5	3	1	0
	Opp10-22	1	11	3	1	0
	Opp22	2	16	10	2	0
		6 (7%)	38 (47%)	22 (27%)	7 (9%)	8 (10%)



	Ground Passes											
	Wasps		Leicester		Bath		Bristol		Exeter		Total	
Danny Care	31	57%	46	62%	31	51%	48	79%	50	88%	206	67.1%
Martin Landajo	20	37%	23	31%	25	41%	2	3%	0	0%	70	22.8%
Joe Marchant	2	4%			2	3%	3	5%	2	4%	9	2.9%
Scott Baldwin			2	3%					1	2%	3	1.0%
Tyrone Green					1	2%	2	3%	1	2%	4	1.3%
Joe Marler			1	1%					1	2%	2	0.7%
Tom Lawday	1	2%									1	0.3%
Marcus Smith			1	1%							1	0.3%
Matt Symons			1	1%	2	3%	2	3%	0	0%	5	1.6%
Aaron Morris							1	2%	0	0%	1	0.3%
Luke Northmore							1	2%			1	0.3%
Louis Lynagh							1	2%	0	0%	1	0.3%
Joe Gray							1	2%			1	0.3%
Stephen Lewies									1	2%	1	0.3%
Andre Esterhuizen									1	2%	1	0.3%
	54		74		61		61		57		307	